03 · RESULTS & DISCUSSION

Findings & Analysis

A synthesis of usability testing results from 40 respondents, heuristic evaluation findings, and data-driven recommendations for improving the Tigaon – Primary Hospital and Clinic Website.

40	73.1	4.33	2.00
Respondents	SUS Score	Best Item (Q5)	Lowest Item (Q6)

SYSTEM USABILITY SCALE RESULTS

Overall SUS Score

Scores were calculated using the standard SUS formula: odd items → (rating – 1); even items → (5 – rating). The sum of all 10 contributions per respondent is multiplied by 2.5, yielding a final score of 0–100.

73.1

Grade B · Good Usability

Above Industry Average

Exceeds the 68 industry benchmark. Perceived as usable with targeted areas for improvement.

Poor	< 52
Acceptable	52–67
Industry Avg	68
Your Score	73.1
Excellent	>= 85

RAW SURVEY DATA

SUS Response Frequency Table

Number of respondents (out of 40) selecting each rating per SUS statement. Data sourced directly from SurveyResult.xlsx. "Contrib." = average per-respondent SUS contribution per item.

No.	Statement	SD 1	D 2	N 3	A 4	SA 5	Total	Mean	Contrib.
01	I think that I would like to use this system frequently.	1	—	5	24	10	40	4.05	3.050
02	I found the system unnecessarily complex.	2	18	16	2	2	40	2.60	2.400
03	I thought the system was easy to use.	—	—	2	29	9	40	4.17	3.175
04	I think that I would need the support of a technical person.	3	25	10	2	—	40	2.27	2.725
05	I found the various functions in this system were well integrated.	—	—	1	25	14	40	4.33	3.325
06	I thought there was too much inconsistency in this system.	6	28	6	—	—	40	2.00	3.000
07	I would imagine that most people would learn to use this system quickly.	—	—	5	24	11	40	4.15	3.150
08	I found the system very cumbersome to use.	7	27	4	2	—	40	2.02	2.975
09	I felt very confident using the system.	—	—	6	26	8	40	4.05	3.050
10	I needed to learn a lot of things before I could get going.	3	20	10	4	3	40	2.60	2.400
								Σ × 2.5 =	73.1

Note: SD = Strongly Disagree, D = Disagree, N = Neutral, A = Agree, SA = Strongly Agree. "—" = zero responses. Positive items (Q1,3,5,7,9): Contrib = Mean – 1. Negative items (Q2,4,6,8,10): Contrib = 5 – Mean.

HEURISTIC EVALUATION

Nielsen's 10 Heuristics — Violation Summary

All 4 group members independently evaluated the prototype using Jakob Nielsen's 10 Usability Heuristics. The table below aggregates violations found across all evaluators.

#	Heuristic	Violations	Severity
H1	Visibility of System Status	2	Medium
H2	Match Between System and Real World	1	Low
H3	User Control and Freedom	3	High
H4	Consistency and Standards	2	Medium
H5	Error Prevention	2	Medium
H6	Recognition Rather Than Recall	1	Low
H7	Flexibility and Efficiency of Use	2	Medium
H8	Aesthetic and Minimalist Design	3	High
H9	Help Users Recognize & Recover from Errors	1	Low
H10	Help and Documentation	2	Medium

KEY ISSUES IDENTIFIED

Usability Problems

Issues identified through heuristic evaluation and open-ended feedback from 40 respondents, cross-referenced with low-scoring SUS items. Sorted by severity.

No Feedback on Form Submission

HIGH

After submitting forms, users received no visible confirmation, leaving them uncertain whether submissions were received. Open-ended responses corroborated this. Aligns with H1 violations identified by all 4 evaluators.

`Violates H1: Visibility of System Status · H9: Help Users Recover from Errors`

Cluttered Homepage Layout

HIGH

The homepage presents too many competing elements, burying the primary call-to-action. Despite Q5 scoring highest (avg 4.33 — functions well integrated), discoverability remains a barrier — consistent with H8 violations flagged by all evaluators.

`Violates H8: Aesthetic and Minimalist Design · H3: User Control and Freedom`

Insufficient Doctor/Specialist Directory

MEDIUM

Respondents noted the specialist directory lacked availability information and role details, making it feel unhelpful. Reflects an H2 violation where the system does not match real-world hospital directory expectations.

`Violates H2: Match Between System and Real World · H6: Recognition Rather Than Recall`

DISCUSSION OF FINDINGS

What the Data Tells Us

An interpretation of usability test results in the context of HCI principles and real respondent feedback.

The average SUS score of **73.1** places the Tigaon – Primary Hospital and Clinic Website in the **"Good" usability range (Grade B)**, exceeding the industry benchmark of 68. This indicates the system is generally usable for its intended audience — 40 respondents predominantly composed of 2nd-year education students aged 19–26 from Partido State University.

The highest-performing items were **Q5** ("Functions well integrated," avg 4.33), **Q3** ("Easy to use," avg 4.18), and **Q7** ("Most people would learn quickly," avg 4.15). These confirm that the system's core features work coherently and are perceived as learnable — genuine design strengths to preserve.

Negatively worded items also performed well: **Q6** ("Too much inconsistency," avg 2.00) and **Q8** ("Cumbersome to use," avg 2.03) both had very low averages — meaning 28 and 27 of 40 respondents respectively *disagreed* that the system was inconsistent or cumbersome. This is a significant positive finding.

Despite the positive SUS outcome, **Q10** ("Needed to learn a lot," avg 2.60) and **Q2** ("Unnecessarily complex," avg 2.60) showed the widest spread in responses — 7 and 4 respondents respectively agreeing with these negative statements. This reveals a learnability gap for a subset of users, particularly the 20 of 40 respondents who had no prior experience with similar systems.

Qualitative open-ended responses significantly enriched the quantitative findings. The most common theme was the need for more detailed specialist/doctor information and better booking confirmation feedback.

The evaluation reveals a system with a strong functional foundation — but with specific, highly actionable gaps in feedback mechanisms, onboarding, and information depth. Addressing these would realistically push the SUS score into the Excellent range (≥ 85).

RECOMMENDATIONS

Suggested Design Improvements

Based on SUS data from 40 respondents, open-ended qualitative feedback, and heuristic evaluation findings — ranked by priority.

01

Implement Visible Form Submission Feedback

Add clear success/error toast notifications or confirmation screens upon form submission. Users should immediately know whether their appointment or inquiry was received. This resolves H1 and H9 violations and should improve Q9 ("felt confident") scores in future evaluations.

02

Enrich the Doctor and Specialist Directory

Expand each doctor profile to include specialization details, consultation schedules, and real-time availability. Respondents noted the current directory lacked actionable information. This directly addresses H2 violations and the most commonly cited open-ended feedback issue from the 40 respondents.

03

Redesign Homepage with a Clear Visual Hierarchy

Simplify the homepage by placing the primary call-to-action ("Book Appointment") above the fold and restructuring the service directory into scannable cards. This resolves H8 violations and will improve immediate usability for first-time visitors — directly targeting the learnability gap in Q10 (avg 2.60).